

PROJECT PROPOSAL

Project Title: Line Following Robot Without Microcontroller

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Objective:

To design and implement a line-following robot that can follow a black line on a white surface using simple electronic circuits instead of a microcontroller. The project demonstrates basic automation using sensors and comparators.

Components Required:

- Infrared (IR) sensors (2 units)
- Comparator IC (LM324 / LM358)
- Transistors (BC547 / BC557)
- Resistors and Potentiometers
- Diodes and LEDs
- DC Motors (2 units)
- Motor Driver Circuit (L293D or relay-based)
- Power supply (9V/12V battery)
- Chassis with wheels

Working Principle:

The robot uses IR sensors to detect a black line. Sensor signals are processed by a comparator circuit to control the motors:

- Left sensor detects black -> turn left
- Right sensor detects black -> turn right
- Both sensors on white -> move straight

This allows the robot to follow the line automatically without a microcontroller.

Expected Output:

A functional line-following robot that automatically detects and follows a black line, adjusting its direction according to sensor input.

Conclusion:

This project demonstrates that a line-following robot can operate without a microcontroller, using only sensors, comparators, and motor drivers. It is low-cost, easy to build, and ideal for learning basic robotics concepts.